Pavlo Radiuk

Associate Professor · Ph.D. in Computer Science
Department of Computer Science, Faculty of Information Technologies, Khmelnytskyi National University, Ukraine

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I work on explainable artificial intelligence. My methods map what a deep network has learned onto the features that clinicians and engineers already use, as a transition matrix between the two feature spaces, so that its decisions can be checked and audited. I validate them against expert judgement in cardiac MRI, ECG and chest radiography, and apply them to UAV inspection of wind and solar plants and to urban traffic control. I am a Researcher in the Khmelnytskyi traffic pilot of the Horizon Europe project U_CAN at Khmelnytskyi National University, and a full-time doctoral candidate for the Doctor of Science degree.

Research interests

Explainable AI	Transition matrices between a model's feature space and an expert's, extended with Lie-group equivariance, rough-set production rules and conflict-aware auditing.
Medical AI	Explanations for ECG, cardiac MRI and chest radiographs, evaluated against clinical expert judgement.
Energy and UAV inspection	Detection and criticality assessment of defects in wind turbines and photovoltaic plants from UAV imagery.
Smart mobility	Reinforcement-learning signal control and traffic-pattern recognition for urban mobility.

Academic and research positions

Sep 2025 – present — **Associate Professor**
Department of Computer Science, Faculty of Information Technologies, Khmelnytskyi National University, Khmelnytskyi, Ukraine
Teaches eight disciplines at bachelor's, master's and Ph.D. levels, including a master's course taught in English, alongside full-time doctoral research for the Doctor of Science degree.

May 2024 – present — **Researcher, Horizon Europe project U_CAN**
Khmelnytskyi National University, Khmelnytskyi, Ukraine
Ukraine towards Carbon Neutrality (grant agreement 101148374), a Coordination and Support Action under the EU Mission for Climate-Neutral and Smart Cities.
- Researcher in the Khmelnytskyi traffic pilot (simulation-based reinforcement-learning signal control)

Dec 2022 – 2024 — **Associate Professor (part-time)**
International Humanitarian University (now International University), Odesa, Ukraine
Taught Machine Learning at master's level and System Analysis at bachelor's level.

Sep 2022 – Aug 2025 — **Senior Lecturer**
Department of Computer Science, Khmelnytskyi National University, Khmelnytskyi, Ukraine

Sep 2021 – Aug 2022 — **Assistant**
Department of Computer Science, Khmelnytskyi National University, Khmelnytskyi, Ukraine

Jan 2021 – Aug 2021 — **Trainee Lecturer**
Department of Computer Science and Information Technologies, Khmelnytskyi National University, Khmelnytskyi, Ukraine

Sep 2017 – Jun 2018 — **Teaching Assistant (hourly)**
Khmelnytskyi National University, Khmelnytskyi, Ukraine

Dec 2016 – Aug 2017 — **Senior Laboratory Assistant**
Department of Applied Mathematics and Social Informatics, Khmelnytskyi National University, Khmelnytskyi, Ukraine

Industry experience

Sep 2022 – present
Data Scientist (advisory)
Horizon Dynamics, Khmelnytskyi, Ukraine
- Research and modelling for data-driven client projects
- Capability-gap assessment and backlog prioritisation in computer vision
- Definition and maintenance of evaluation metrics for ongoing projects

Sep 2022 – Oct 2023
Data Scientist
International e-commerce client (via Nolt Technologies), Khmelnytskyi, Ukraine
Remote engagement delivered through Nolt Technologies.
- NLP recommendation models (sequential transformers) for personalised product recommendations
- Text-classification models (LSTM, BERT-family) for client segmentation
- Sales-forecasting and cash-flow prediction models on AWS

2019 – 2020
Business & Data Analyst
MEDICA.TEC, Khmelnytskyi, Ukraine
Business and data analysis for a medical-technology company.

Mar 2018 – Apr 2019
Data Analyst / Business Analyst
NGO ASTAR (Sustainable Development Agency ASTAR), Khmelnytskyi, Ukraine
- Commissioned market-analysis reports for regional producers
- Business audit and translation of business objectives into technical requirements
- Recurring statistical reporting

Education

2025 – 2027 (expected)
D.Sc. (Doctor of Science, habilitation level), Information Technologies
Khmelnytskyi National University, Khmelnytskyi
Thesis: *Theoretical and Applied Foundations of an Intelligent Information Technology for Interpreting Decisions Made by Artificial Intelligence Systems*
Prof. Oleksandr Barmak (scientific consultant)

2022
Ph.D., Computer Science
Khmelnytskyi National University, Khmelnytskyi
Thesis: *Information Technology for the Early Diagnosis of Pneumonia by Individual Selection of the Parameters of a Classification Model for Medical Images of the Lungs*
Supervisor: Prof. Oleksandr Barmak

2017
M.Sc., Mathematical and Computer Modelling
Khmelnytskyi National University, Khmelnytskyi, with distinction
Thesis: *Optimisation of Filter Size Distribution in Convolutional Neural Networks for Classification Problems*

2015
B.Sc., Applied Mathematics
Khmelnytskyi National University, Khmelnytskyi

Research projects

May 2024 – Apr 2028 **Ukraine towards Carbon Neutrality (U_CAN)**
Horizon Europe, EU Mission for Climate-Neutral and Smart Cities, grant agreement 101148374
Role: Researcher, Khmelnytskyi traffic pilot
A Coordination and Support Action coordinated by Technische Universität Dresden that supports Ukrainian cities on their path to climate neutrality. Khmelnytskyi National University leads the Khmelnytskyi pilot on adaptive traffic management, which develops traffic-pattern recognition and reinforcement-learning signal control and tests them in a calibrated simulation of the city's road network.



Funded by the European Union. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Climate, Infrastructure and Environment Executive Agency (CINEA). Neither the European Union nor the granting authority can be held responsible for them.

Sep 2024 – Aug 2026 **Intelligent System for Recognising Defects of Green-Energy Objects Using UAVs**
Ministry of Education and Science of Ukraine, state-funded research and development, state registration No. 0124U004665
Role: Researcher (executor)
A national project led by West Ukrainian National University with the participation of Khmelnytskyi National University. It develops methods for collecting, recognising and processing UAV imagery to detect defects in wind-energy installations; its published outputs extend to solar plants.

2021 – 2022 **Development of Information Technology for Making Human-Controlled Critical and Safety Decisions Based on Mental-Formal Models of Machine Learning**
Ministry of Education and Science of Ukraine, state-funded research and development, state registration No. 0121U112025
Role: Researcher (executor)
An applied-research project of Khmelnytskyi National University led by Prof. Oleksandr Barmak. It developed an architecture and machine-learning methods for safety-critical decisions made with artificial intelligence, including a data-processing system that shows which factors drive each decision and how to interpret it.

2021 – 2022 **Self-Organized Distributed System for Malware Detection in Computer Networks**
Ministry of Education and Science of Ukraine, state-funded research and development, state registration No. 0121U109936
Role: Researcher (executor)
An applied-research project of Khmelnytskyi National University led by Prof. Tetiana Hovorushchenko. It developed the theoretical framework for, and built, a self-organised distributed system that detects malware in computer networks.

Awards

2022–2024 **Scholarship of the Cabinet of Ministers of Ukraine for Young Scientists**
Committee on State Prizes of Ukraine in Science and Technology
National competitive award; Resolution No. 6 of 4 November 2022.

2023 **Best Paper Award for Ukrainian authors**
12th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems (IDAACS-2023)
Awarded in Dortmund, Germany, on 9 September 2023 for the paper "Apple detection with occlusions using modified YOLOv5-v1" (O. Melnychenko, O. Savenko, P. Radiuk).

Service

Peer review	450+ reviews for journals and conferences (Web of Science).
Programme committees	International Conference on Database and Expert Systems Applications (DEXA), 2024–2026 International Conference on Web Information Systems Engineering (WISE), 2024, 2026 MobiWis, 2026
Editorial	Guest Editor, Sustainability (MDPI), 2024–2026 Volume Editor, CEUR Workshop Proceedings, Vol. 4141 (ExplAI-2025), 2025 Editorial Board Member and Handling Editor, Journal of Intelligent & Fuzzy Systems, 2023–2024
Organising	Organising Committee member, Workshop on Advanced AI in Explainability and Ethics for the Sustainable Development Goals (ExplAI), 2025–2026 Lead organiser, International conference "Reducing CO_2 Emissions in Ukrainian Cities: Pathways to Climate Neutrality" (Horizon Europe U_CAN), Khmelnytskyi National University, 2026
Doctoral defences	Reviewer, two one-off specialised academic councils for Ph.D. defences, Khmelnytskyi National University, 2026 Official Opponent, Ph.D. defence, Ternopil Ivan Puluj National Technical University, 2026
Conference reviewing	International Workshop on Intelligent Information Technologies & Systems of Information Security (IntelITSIS), 2022–2026; YAISD, 2025–2026; IEEE International Conference on Advanced Computer Information Technologies (ACIT), 2026; APWeb-WAIM, 2026; International Workshop on Intelligent & CyberPhysical Systems (ICyberPhyS), 2024–2025; IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS), 2025; International Scientific Symposium "Intelligent Solutions" (IntSol), 2025; International Conference on Web Information Systems Engineering (WISE), 2025; International Conference on Informatics & Data-Driven Medicine (IDDM), 2022–2024; International Workshop on Advanced Applied Information Technologies (AdvAIT), 2024; ICST, 2024.

Teaching

Ph.D.	Project Management in Research and Development; Modern Methodologies, Methods and Tools of Experimental and Theoretical Research in Computer Science.
Master's	Conducting Experiments and Processing the Results of Scientific Research (in English) (open course materials).
Bachelor's	Decision Theory and Operations Research; Intelligent Data Analysis (open course materials); Operations Research and Fundamentals of Decision Theory (open course materials); Probability Theory and Statistical Data Processing; Statistical Data Processing (open course materials).

Skills and languages

Machine learning	Deep learning · PyTorch · TensorFlow · Keras · Computer vision · NLP and LLMs · Reinforcement learning · scikit-learn · XGBoost · Explainable AI
Engineering	Python · TypeScript · C# · SQL · AWS · Docker · Git · PostgreSQL · MySQL
Analysis and simulation	Statistical modelling · Forecasting · SUMO · PTV Visum · Power BI · Tableau
Languages	Ukrainian — native; English — professional working proficiency (CEFR B2 certified; teaches and publishes in English).

Collaboration

I welcome partners for Horizon Europe, MSCA and EuroHPC proposals in these roles.

XAI methods lead	Task or work-package lead for explainable and trustworthy AI: transition-matrix explanations, rough-set rule extraction and audit protocols that domain experts can check.
Medical AI partner	Explainable deep learning for ECG, cardiac MRI and chest radiographs. Explanations for ECG and cardiac MRI have been evaluated against clinical expert judgement, with Cohen's kappa of 0.89 and 0.80 respectively.
Energy & UAV inspection	UAV-based detection and criticality assessment of defects in wind turbines and photovoltaic modules from RGB, thermal and multispectral imagery, developed with Ukrainian and Polish partners.
Ukrainian pilot site	Khmelnytskyi National University as a pilot site in Ukraine, drawing on the U_CAN Khmelnytskyi traffic pilot: a calibrated city-scale traffic simulation, joint training with the city council and an international conference organised in 2026.
Organisation	Khmelnytskyi National University, Participant Identification Code (PIC) 964860456. Capability statement, biographies and photograph: radiukpavlo.github.io/partner-kit.

Publications

76 works: 38 journal articles, 34 conference papers, 2 book chapters and 2 preprints. Own name in bold; † marks a mentored junior researcher. Works link to their DOI or publisher page where one exists, with open-access PDFs and code where available; the list with BibTeX is at radiukpavlo.github.io/publications.

Citation indicators each database counts its own coverage, so figures are not summed

Database	Documents	Citations	h-index	i10-index	As of
Google Scholar	–	1,158	17	24	14 Oct 2026
Scopus	53	333	10	–	12 Oct 2026
Web of Science Core Collection	25	131	7	–	12 Oct 2026

Journal articles 38

J38 **A Geo-consistent Weather Augmentation Method for Improving UAV Localization Robustness under Adverse Meteorological Conditions**
V. Vozniak†, O. Barmak, **P. Radiuk**, I. Krak.
Journal of Intelligent & Robotic Systems, 2026. doi:10.1007/s10846-026-02454-1

J37 **A method for sentiment analysis of customer reviews based on hybrid classification models for automating feedback in IT project management**
B. Zahurskyi, E. Manziuk, T. Skrypnyk, **P. Radiuk**.
Measuring and Computing Devices in Technological Processes No. 3, pp. 487–496, 2026 (in Ukrainian). doi:10.31891/2219-9365-2026-87-56 · PDF

J36 **Architecture of cyberphysical systems for UAV-based late-fusion defect detection in photovoltaic modules**
A. Sachenko, **P. Radiuk**, M. Lysyi, O. Melnychenko, O. Zastavnyy.
Innovative Technologies and Scientific Solutions for Industries No. 1(35), pp. 79–99, 2026. doi:10.30837/2522-9818.2026.1.079 · PDF

J35 **Automated Synthesis of Hierarchical Deep Learning Cascades for Identifying Visually Similar Objects in UAV Imagery**
D. Borovyk†, O. Barmak, **P. Radiuk**, I. Krak.
Technologies 14(6):360, 2026. doi:10.3390/technologies14060360 · PDF

J34 **CARDIA-X: Global Semantic Transition and Rough-Set Rules for Auditable Post Hoc Electrocardiographic Explainability**
P. Radiuk, O. Barmak, L. Klymenko, I. Krak.
AI 7(9):376, 2026. doi:10.3390/ai7090376 · PDF · Code

J33 **Equivariant Transition Matrices for Explainable Deep Learning: A Lie Group Linearization Approach**
P. Radiuk, O. Barmak, L. Bedratyuk, I. Krak.
Machine Learning and Knowledge Extraction 8(4):92, 2026. doi:10.3390/make8040092 · PDF · Code

J32 **SEMTRA: Global Semantic Transition and Rough-Set Rules for Auditable Post-Hoc Explainability**
P. Radiuk, O. Barmak, I. Krak.
Machine Learning and Knowledge Extraction 8(7):181, 2026. doi:10.3390/make8070181 · PDF · Code

J31 **Topography-Aware Deep Reinforcement Learning with Contextual Reward Engineering for Sustainable and Efficient Urban Traffic Control**
O. Ryzhanskyi[†], O. Barmak, E. Manziuk, **P. Radiuk**, I. Krak.
Future Transportation 6(2):82, 2026. doi:10.3390/futuretransp6020082 · PDF

J30 **WEDD-RST: An interpretive approach to detecting defects in photovoltaic panels in cyber-physical systems**
P. Radiuk, A. Sachenko, O. Melnychenko, R. Brukhanskyi, A. Kashtalian.
Innovative Technologies and Scientific Solutions for Industries No. 2(36), pp. 153–172, 2026. doi:10.30837/2522-9818.2026.2.153 · PDF

J29 **A method for interpreting fake news detection results using a large language model**
S. Vovk, **P. Radiuk**, T. Skrypnyk.
Herald of Khmelnytskyi National University. Technical Sciences 359(6.2), pp. 149–156, 2025 (in Ukrainian). doi:10.31891/2307-5732-2025-359-91 · PDF

J28 **An Adaptive Machine Learning Approach to Sustainable Traffic Planning: High-Fidelity Pattern Recognition in Smart Transportation Systems**
V. Pavlyshyn[†], E. Manziuk, O. Barmak, **P. Radiuk**, I. Krak.
Future Transportation 5(4):152, 2025. doi:10.3390/futuretransp5040152 · PDF

J27 **Criticality Assessment of Wind Turbine Defects via Multispectral UAV Fusion and Fuzzy Logic**
P. Radiuk, B. Rusyn, O. Melnychenko, T. Perzynski, A. Sachenko, S. Svystun[†], O. Savenko.
Energies 18(17):4523, 2025. doi:10.3390/en18174523 · PDF

J26 **DyTAM: Accelerating Wind Turbine Inspections with Dynamic UAV Trajectory Adaptation**
S. Svystun[†], L. Scislo, M. Pawlik, O. Melnychenko, **P. Radiuk**, O. Savenko, A. Sachenko.
Energies 18(7):1823, 2025. doi:10.3390/en18071823 · PDF

J25 **Enhanced fire hazard detection in solar power plants: an integrated UAV, AI, and SCADA-based approach**
A. Lysyi, A. Sachenko, **P. Radiuk**, M. Lysyi, O. Melnychenko, O. Ishchuk, O. Savenko.
Radioelectronic and Computer Systems 2025(2), pp. 99–117, 2025. doi:10.32620/reks.2025.2.06

J24 **Hierarchical Deep Learning Model for Identifying Similar Targets in UAV Imagery**
D. Borovyk[†], O. Barmak, **P. Radiuk**, I. Krak.
Drones 9(11):743, 2025. doi:10.3390/drones9110743 · PDF

J23 **Method of adaptive knowledge distillation from multi-teacher to student deep learning models**
O. Chaban[†], E. Manziuk, **P. Radiuk**.
Journal of Edge Computing 4(2), pp. 159–178, 2025. doi:10.55056/jec.978 · PDF

J22 **Method of UAV-based inspection of photovoltaic modules using thermal and RGB data fusion**
A. Lysyi, A. Sachenko, **P. Radiuk**, M. Lysyi, O. Melnychenko, D. Zahorodnia.
Radioelectronic and Computer Systems 2025(4), pp. 186–205, 2025. doi:10.32620/reks.2025.4.13

J21 **Towards Transparent AI in Medicine: ECG-Based Arrhythmia Detection with Explainable Deep Learning**
O. Kovalchuk[†], O. Barmak, **P. Radiuk**, L. Klymenko, I. Krak.
Technologies 13(1):34, 2025. doi:10.3390/technologies13010034 · PDF

J20 **Unsupervised Knowledge Extraction of Distinctive Landmarks from Earth Imagery Using Deep Feature Outliers for Robust UAV Geo-Localization**
Z. Ostrovskyi[†], O. Barmak, **P. Radiuk**, I. Krak.
Machine Learning and Knowledge Extraction 7(3):81, 2025. doi:10.3390/make7030081 · PDF

J19 **Explainable Deep Learning: A Visual Analytics Approach with Transition Matrices**
P. Radiuk, O. Barmak, E. Manziuk, I. Krak.
Mathematics 12(7):1024, 2024. doi:10.3390/math12071024 · PDF · Code

J18 **Intelligent Integrated System for Fruit Detection Using Multi-UAV Imaging and Deep Learning**
O. Melnychenko, L. Scislo, O. Savenko, A. Sachenko, **P. Radiuk**.
Sensors 24(6):1913, 2024. doi:10.3390/s24061913 · PDF

J17 **Thermal and RGB Images Work Better Together in Wind Turbine Damage Detection**
S. Svystun[†], O. Melnychenko, **P. Radiuk**, O. Savenko, A. Sachenko, A. Lysyi.
International Journal of Computing 23(4), pp. 526–535, 2024. doi:10.47839/ijc.23.4.3752

J16 **Toward explainable deep learning in healthcare through transition matrix and user-friendly features**
O. Barmak, I. Krak, S. Yakovlev, E. Manziuk, **P. Radiuk**, V. Kuznetsov.
Frontiers in Artificial Intelligence 7:1482141, 2024. doi:10.3389/frai.2024.1482141 · PDF

J15 **Facial Emotion Recognition for Photo and Video Surveillance Based on Machine Learning and Visual Analytics**
O. Kalyta†, O. Barmak, **P. Radiuk**, I. Krak.
Applied Sciences 13(17):9890, 2023. doi:10.3390/app13179890 · PDF

J14 **Information system for public places and institutions visualization with opportunities of inclusive access and optimal routing**
O. Pavlova, **P. Radiuk**, S. Kravchuk, V. Kulbachnyi.
Computer systems and information technologies No. 1, pp. 62–68, 2022. doi:10.31891/CSIT-2022-1-8

J13 **Semantic alignment of ontologies meaningful categories with the generalization of descriptive structures**
E. Manziuk, O. Barmak, I. Krak, O. A. Pasichnyk, **P. Radiuk**, O. V. Mazurets.
Problems in Programming No. 3-4, pp. 355–363, 2022. doi:10.15407/pp2022.03-04.355

J12 **An Approach to Early Diagnosis of Pneumonia on Individual Radiographs based on the CNN Information Technology**
P. Radiuk, O. Barmak, I. Krak.
The Open Bioinformatics Journal 14(1), pp. 93–107, 2021. doi:10.2174/1875036202114010093 · PDF · Code

J11 **Applying mental models to making controlled critically safe decisions in IT project management**
P. Radiuk, T. Skrypnyk, D. Karlechuk.
Herald of Khmelnytskyi National University. Technical Sciences 301(5), pp. 32–35, 2021. doi:10.31891/2307-5732-2021-301-5-32-35

J10 **Approaches to practical analysis of computing algorithms**
O. Barmak, **P. Radiuk**, M. Molchanova, O. Sobko.
Herald of Khmelnytskyi National University. Technical Sciences 303(6), pp. 102–105, 2021 (in Ukrainian). doi:10.31891/2307-5732-2021-303-6-102-105

J9 **Information technology for early diagnosis of pneumonia using convolutional neural networks**
P. Radiuk, O. Barmak, I. Krak.
Problems of Control and Informatics 66(3), pp. 101–114, 2021 (in Ukrainian). doi:10.34229/1028-0979-2021-3-9 · PDF

J8 **Information technology of visual analysis of X-ray images for interpretation of pneumonia diagnostic results**
O. Barmak, **P. Radiuk**.
Herald of Khmelnytskyi National University. Technical Sciences 295(2), pp. 52–55, 2021 (in Ukrainian). doi:10.31891/2307-5732-2021-295-2-52-55

J7 **Intelligent data analysis using artificial neural networks for decision making in the education domain**
P. Radiuk, O. V. Mazurets, T. Skrypnyk, O. Moroz.
Herald of Khmelnytskyi National University. Technical Sciences 303(6), pp. 111–114, 2021. doi:10.31891/2307-5732-2021-303-6-111-114

J6 **Web-based information technology for classifying and interpreting early pneumonia based on fine-tuned convolutional neural network**
P. Radiuk, O. Barmak.
Computer systems and information technologies No. 1, pp. 12–18, 2021. doi:10.31891/CSIT-2021-3-2

J5 **An approach to accelerate the training of convolutional neural networks by tuning the hyperparameters of learning**
P. Radiuk.
Computer Systems and Information Technologies 2(2), pp. 32–37, 2020 (in Ukrainian). doi:10.31891/CSIT-2020-2-5

J4 **Application of a genetic algorithm to search for the optimal convolutional neural network architecture with weight distribution**
P. Radiuk.
Herald of Khmelnytskyi National University. Technical Sciences 281(1), pp. 7–11, 2020. doi:10.31891/2307-5732-2020-281-1-7-11

J3 **Applying 3D U-Net Architecture to the Task of Multi-Organ Segmentation in Computed Tomography**
P. Radiuk.
Applied Computer Systems 25(1), pp. 43–50, 2020. doi:10.2478/acss-2020-0005

J2 **Neuroevolution of convolutional neural networks for the classification of lung cancer images**
P. Radiuk.
Herald of Khmelnytskyi National University. Technical Sciences 267(6(2)), pp. 188–192, 2018.

J1 **Impact of Training Set Batch Size on the Performance of Convolutional Neural Networks for Diverse Datasets**
P. Radiuk.
Information Technology and Management Science 20(1), pp. 20–24, 2017. doi:10.1515/itms-2017-0003

Conference papers 34

C34 **Intelligent information system for knowledge integration into artificial intelligence models**
O. Chaban†, E. Manziuk, **P. Radiuk**, E. Zaitseva, O. Markevych.
International Workshop on Advanced Applied Information Technologies: AI & DSS 2025, CEUR Workshop Proceedings 4163, pp. 145–161, 2026. ceur-ws.org · PDF

C33 **AI-Driven Traffic Signal Control System to Reduce CO2 Emissions**
O. Ryzhanskyi†, V. Pavlyshyn†, **P. Radiuk**, E. Manziuk, O. Barmak, I. Krak.
YAISD Workshops 2025, CEUR Workshop Proceedings 3974, pp. 18–27, 2025. ceur-ws.org · PDF

C32 **Determining the criticality assessment of defects on wind turbine components detected by UAV sensors**
S. Svystun†, O. Melnychenko, **P. Radiuk**, A. Lysyi, A. Sachenko.
Intelligent Information Technologies & Systems of Information Security 2025, CEUR Workshop Proceedings 3963, pp. 351–362, 2025. ceur-ws.org · PDF

C31 **Distributed intelligent system architecture for UAV-Assisted monitoring of wind energy infrastructure**
S. Svystun†, O. Melnychenko, **P. Radiuk**, O. Savenko, A. Lysyi.
The 1st International Workshop on Advanced Applied Information Technologies, CEUR Workshop Proceedings 3899, pp. 150–161, 2025. ceur-ws.org · PDF

C30 **EMTKD at the edge: An adaptive multi-teacher knowledge distillation for robust cardiac MRI classification**
O. Chaban†, E. Manziuk, O. Markevych, S. Petrovskyi, **P. Radiuk**.
5th Edge Computing Workshop, CEUR Workshop Proceedings 3943, pp. 42–57, 2025. ceur-ws.org · PDF

C29 **Establishing Patterns of the Urban Transport Flows on Clustering Analysis**
V. Pavlyshyn†, O. Ryzhanskyi†, E. Manziuk, **P. Radiuk**, O. Barmak, I. Krak.
YAISD Workshops 2025, CEUR Workshop Proceedings 3974, pp. 1–9, 2025. ceur-ws.org · PDF

C28 **Feature Computation Procedure for Fake News Detection: An LLM-Based Extraction Approach**
A. Shupta†, **P. Radiuk**, I. Krak.
Intelligent Information Technologies & Systems of Information Security 2025, CEUR Workshop Proceedings 3963, pp. 112–124, 2025. ceur-ws.org · PDF

C27 **Multi-stage Segmentation and Cascade Classification Methods for Improving Cardiac MRI Analysis**
V. Slobodzian†, **P. Radiuk**, O. Barmak, I. Krak.
Information Technology and Implementation 2024, CEUR Workshop Proceedings 3909, pp. 84–98, 2025. ceur-ws.org · PDF

C26 **Verifiable by construction: Evidence-anchored LLMs for explainable fake news detection**
A. Shupta†, **P. Radiuk**, M. Kvassay, P. Gaj.
Advanced AI in Explainability and Ethics for the Sustainable Development Goals, CEUR Workshop Proceedings 4141, pp. 168–182, 2025. ceur-ws.org · PDF

C25 **YOLO Ensemble for UAV-based Multispectral Defect Detection in Wind Turbine Components**
S. Svystun†, **P. Radiuk**, O. Melnychenko, O. Savenko, A. Sachenko.
2025 IEEE 13th International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS), pp. 11–16, 2025. doi:10.1109/IDAACS68557.2025.11322062

C24 **Dynamic Trajectory Adaptation for Efficient UAV Inspections of Wind Energy Units**
S. Svystun†, O. Melnychenko, **P. Radiuk**, O. Savenko, A. Sachenko, A. Lysyi.
2024 14th International Conference on Dependable Systems, Services and Technologies (DESSERT), pp. 1–7, 2024. doi:10.1109/DESSERT65323.2024.11122243

C23 **ECG Arrhythmia Classification and Interpretation using Convolutional Networks for Intelligent IoT Healthcare System**
O. Kovalchuk†, O. Barmak, **P. Radiuk**, I. Krak.
Intelligent & CyberPhysical Systems 2024, CEUR Workshop Proceedings 3736, pp. 47–62, 2024. ceur-ws.org · PDF

C22 **Explainable Artificial Intelligence: Transitioning DL Model Decisions to User-Understandable Features in Healthcare**
P. Radiuk, O. Barmak, E. Manziuk, I. Krak.
4th International Workshop of IT-professionals on Artificial Intelligence 2024, CEUR Workshop Proceedings 3777, pp. 185–199, 2024. ceur-ws.org · PDF

C21 **Integration of Contextual Descriptors in Ontology Alignment for Enrichment of Semantic Correspondence**
E. Manziuk, O. Barmak, **P. Radiuk**, V. Kuznetsov, I. Krak, S. Yakovlev.
2024 14th International Conference on Dependable Systems, Services and Technologies (DESSERT), pp. 1–7, 2024. doi:10.1109/DESSERT65323.2024.11122179

C20 **Interpretable deep learning method for medical image diagnosis**
E. Manziuk, O. Barmak, I. Krak, N. Petliak, Z. Jin, **P. Radiuk**.
Proceedings of the International Scientific Conference Intellectual Systems of Decision Making and Problems of Computational Intelligence (ISDMCI'2024), pp. 26–28, 2024.

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Presentations

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Oct 2026 Intelligent Transport Systems: Ecology, Safety, Quality, Comfort (ITS ESQC 2026), Kyiv, Ukraine.
Adaptive cascade clustering for pattern-aware traffic-regime identification at locally constrained urban intersections

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Sep 2026 Information Control Systems and Technologies (ICST-ODESA 2026), Odesa, Ukraine.
Auditing deep learning for cardiac MRI through semantic transitions and conflict-aware production rules

Sep 2026 U_CAN Scenario Stress-Testing Workshop, TU Dresden (U_CAN Dresden 2026), Dresden, Germany.
Khmelnytskyi National University input on the Khmelnytskyi adaptive-traffic pilot

Jul 2026 U_CAN consortium meeting (U_CAN L'Aquila 2026), L'Aquila, Italy.
Work package 4 status presentation

Apr 2026 International conference "Reducing CO_2 Emissions in Ukrainian Cities: Pathways to Climate Neutrality" (Horizon Europe U_CAN) (U_CAN conference 2026), Khmelnytskyi, Ukraine.
Lead organiser and speaker

Dec 2025 2nd International Workshop on Advanced Applied Information Technologies: AI & DSS (AdvAIT 2025), Khmelnytskyi, Ukraine.
Intelligent information system for knowledge integration into artificial intelligence models

Nov 2025 1st Workshop on Advanced AI in Explainability and Ethics for the Sustainable Development Goals (ExplAI-2025), Khmelnytskyi, Ukraine.
Verifiable by construction: Evidence-anchored LLMs for explainable fake news detection

Sep 2025 13th IEEE International Conference on Intelligent Data Acquisition and Advanced Computing Systems: Technology and Applications (IDAACS 2025), Gliwice, Poland.
YOLO ensemble for UAV-based multispectral defect detection in wind turbine components

May 2025 Joint workshops AI for Environmental and Social Sustainability and AI and Interdisciplinary Innovations for Sustainable Development (YAISD-WS 2025), Ternopil, Ukraine.
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Apr 2025 6th International Workshop on Intelligent Information Technologies & Systems of Information Security (IntelITSIS 2025), Khmelnytskyi, Ukraine.
Feature computation procedure for fake news detection: An LLM-based extraction approach

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Distributed intelligent system architecture for UAV-assisted monitoring of wind energy infrastructure

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Multi-stage segmentation and cascade classification methods for improving cardiac MRI analysis

Oct 2024 14th International Conference on Dependable Systems, Services and Technologies (DESSERT 2024), Athens, Greece.
Integration of contextual descriptors in ontology alignment for enrichment of semantic correspondence

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Explainable artificial intelligence: Transitioning DL model decisions to user-understandable features in healthcare

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ECG arrhythmia classification and interpretation using convolutional networks for intelligent IoT healthcare system

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Heuristic architecture search using network morphism for chest X-ray classification

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